

EXERCISE 1 - CREATE A USER

--1.1

```
CREATE USER c##Student_1701497 IDENTIFIED BY Philbo$$dba;
```

```
GRANT CREATE SESSION to c##Student_1701497;
```

```
GRANT CONNECT to c##Student_1701497;
```

--1.2

```
CONNECT c##Student_1701497/philbo$$dba;
```

--1.3

```
CREATE USER c##test1_1701497 identified by Password1 ACCOUNT LOCK;
```

--1.4

```
/* The user was unable to login and was greeted with "ORA-28000: The
account is locked." */
```

--1.5

```
-- Connect back to the admin account
```

```
CREATE USER c##test2_1701497 identified by Password1 PASSWORD
EXPIRE;
```

```
/*After create user when trying to connect returned an error "ORA-
01045: user C##TEST2_1701497 lacks CREATE SESSION privilege; logon
denied" */
```

```
--Granted session to resolve
```

```
Grant create session to c##test2_1701497;
```

--1.6

```
/*Prompted that password has expire and changing password for
account
```

```
Enter new password and retype for verification purpose
```

Password changed to: Password2*/
EXERCISE 2 - ALTERING USERS

```
Connect c##dbadmin/Password1;

--2.1

Alter user c##test2_1701497 identified by newpassword123 password
expire;

--2.2

Alter USER c##test1_1701497 account UNLOCK;

Alter USER c##test1_1701497 password EXPIRE;

Grant create session to c##test1_1701497;

-- Admin can change password for account by :
Password c##test1_1701497;
```

EXERCISE 3 – DELETING USERS

```
--1

DROP USER c##test1_1701497;

COMMIT;

Select * from dba_users where username = 'c##test1_1701497';

/*Do not see any difference the account has been drop successful
and is not listed in dba users*/
```

EXERCISE 4 – GRANTING PRIVILEGES

```
--1

CONNECT c##Student_1701497/philbo$$dba;

Connect C##student_1701497/philbo$$dba;
```

```
--Create table attempt
Create table Product(
  ID INTEGER,
  Name_Product VARCHAR(25),
  Price FLOAT,
  Description_product VARCHAR(30),
  ExpiryDate DATE);

/* Table was unable to create because of privileges
ORA-01031: insufficient privileges*/

--Granting permission to create table
GRANT CREATE TABLE TO C##student_1701497;

Connect C##student_1701497/philbo$$dba;

Create table Product(
  ID INTEGER,
  Name_Product VARCHAR(25),
  Price FLOAT,
  Description_product VARCHAR(30),
  ExpiryDate DATE);

insert all
into Product VALUES (1001, 'Jimmy Choo', 500.00, 'It is very
nice', TO_DATE('21/02/2023', 'dd/mm/yyyy'))
into Product VALUES (1002, 'Tom Ford', 300.00, 'It is
goldly', TO_DATE('16/04/2023', 'dd/mm/yyyy'))
```

```
into Product VALUES (1003, 'Victoria', 300.00, 'It is very
good', TO_DATE('21/02/2024', 'dd/mm/yyyy'))
SELECT * FROM dual;
```

```
/*ERROR at line 1:
ORA-01950: no privileges on tablespace 'USERS'*/
```

```
Connect sys as sysdba
```

```
--3
```

```
ALTER USER C##student_1701497 QUOTA UNLIMITED ON USERS;
```

```
-- The insert statement ran successfully
```

```
--4
```

```
Connect c##test2_1701497/Password1
```

```
-- Update error ORA-00942: table or view does not exist
```

```
UPDATE Product
SET Price = 950.00
WHERE ID = 1001;
```

```
--Delete
```

```
DELETE FROM Product
WHERE ID = 1003;
```

```
--Insert
```

```
INSERT INTO Product VALUES (1004, 'Supligen', 250.00, 'Get some
energy', TO_DATE('21/02/2025', 'dd/mm/yyyy'));
```

```
--Select
Select * from Product;

--5

CONNECT c##Student_1701497/philbo$$dba;
--Granted the selected privileges to user

GRANT SELECT , INSERT , UPDATE , DELETE ON Product TO
c##test2_1701497;

-- Grant succeeded
Connect c##test2_1701497/Password1;

/*There was problem trying to run query on tables from the
c##test2_1701497
account without entering the username beside the table eg.
[username].[tablename] works*/

--update
UPDATE c##Student_1701497.Product
SET Price = 950.00
WHERE ID = 1001;

--Delete
DELETE FROM c##Student_1701497.Product
WHERE ID = 1003;
```

```
--Insert
INSERT INTO c##Student_1701497.Product VALUES
(1004, 'Supligen', 250.00, 'Get some
energy', TO_DATE('21/02/2025', 'dd/mm/yyyy'));

--Select
Select * from c##Student_1701497.Product;

/*All the statements ran successfully now */

--6

Connect c##dbadmin/Password1;

CREATE USER c##dummy_1701497 IDENTIFIED BY Password1;

GRANT CREATE TABLE TO c##dummy_1701497;
Grant create session to c##dummy_1701497;

Connect c##dummy_1701497/Password1;

--Dummy table
Create table Dummy(
ID INTEGER,
Name_Product VARCHAR(25),
Price FLOAT,
Description_product VARCHAR(30),
ExpiryDate DATE);

DROP USER c##dummy_1701497;
/*ERROR at line 1:
```

```
ORA-01922: CASCADE must be specified to drop 'C##DUMMY_1701497'*/  
/*DROP USER user_name [ CASCADE ]; */
```

```
DROP USER c##dummy_1701497 CASCADE;
```

EXERCISE 5 – CREATING ROLES

```
SHOW USER
```

```
--1
```

```
--A
```

```
CREATE ROLE c##labclassA NOT IDENTIFIED;
```

```
--B
```

```
CREATE ROLE c##labclassB IDENTIFIED BY pass1234;
```

```
--2
```

```
--A
```

```
CREATE USER c##StudentA_1701497 IDENTIFIED BY pass1234;
```

```
--B
```

```
CREATE USER c##StudentB_1701497 IDENTIFIED BY pass1234;
```

```
--3
```

```
Connect c##StudentA_1701497/pass1234;
```

```
/* Lacks create session privilege */
```

```
Connect c##dbadmin/Password1;
```

```
Grant create session to c##StudentA_1701497;
```

```
Connect c##StudentA_1701497/pass1234;
```

```
Create table Sales(  
Sales_Id VARCHAR(25),  
Product VARCHAR(25),  
Unitcost FLOAT,  
quantity INTEGER);  
  
/*ORA-01031: insufficient privileges - table was not created */
```

EXERCISE 6 - SETTING A ROLE

```
--1  
  
Connect c##dbadmin/Password1;  
  
GRANT c##labclassA TO c##StudentA_1701497;  
/*The role is immediately enabled*/  
  
--2  
  
ALTER USER c##StudentA_1701497  
DEFAULT ROLE  
c##labclassA;  
  
/*There wasn't any conflict*/  
  
--3  
GRANT CREATE ANY TABLE TO c##labclassA;  
  
--4  
  
Connect c##StudentA_1701497/pass1234;  
  
Create table Sales(  
Sales_Id VARCHAR(25),  
Product VARCHAR(25),  
Unitcost FLOAT,  
quantity INTEGER);
```



```
/*Table was successfully created*/
```

```
--5
```

```
insert all
  into Sales VALUES ('A001', 'Apple', 20.00, 10)
  into Sales VALUES ('B002', 'Mango', 40.00, 70)
  into Sales VALUES ('C003', 'Pear', 900.00, 20)
SELECT * FROM dual;
```

```
--error: no privileges on tablespace 'Users'
```

```
Connect c##dbadmin/Password1;
```

```
ALTER USER c##StudentA_1701497 QUOTA UNLIMITED ON USERS;
```

```
Connect c##StudentA_1701497/pass1234;
```

```
--Insert values successfully after
```

```
Connect c##StudentA_1701497/pass1234;
```

```
insert all
  into Sales VALUES ('A001', 'Apple', 20.00, 10)
  into Sales VALUES ('B002', 'Mango', 40.00, 70)
  into Sales VALUES ('C003', 'Pear', 900.00, 20)
SELECT * FROM dual;
```

```
--6
```

```
Select * from Sales;
```

```
--a
```

```
Connect c##dbadmin/Password1;
```

```
Grant create session to c##StudentB_1701497;
```

```
Connect c##StudentB_1701497/pass1234;
```

```
Select * from c##StudentA_1701497.Sales;
```

```
-- Table or view does not exist meaning StudentB does not  
has access to sales table as yet
```

```
-- 7
```

```
Connect c##StudentA_1701497/pass1234;
```

```
GRANT SELECT ON Sales TO c##StudentB_1701497
```

```
--Running the select statement from sales as studentB
```

```
Connect c##StudentB_1701497/pass1234;
```

```
Select * from c##StudentA_1701497.Sales;
```

EXERCISE 7

```
--1
```

```
Connect c##dbadmin/Password1;
```

```
GRANT c##labclassB TO c##StudentB_1701497;
```

```
Select *
```

```
FROM DICTIONARY
```

```
WHERE table_name LIKE '%ROLES%';
```

```
-- Helps to understand the column names so easier to narrow  
down a search result
```

```
DESCRIBE DBA_ROLES
```

```
SELECT * FROM DBA_ROLES
```

```
WHERE ROLE LIKE '%C##LABCLASS%'
```

```
--Roles information are stores related tables
```

--2

```
Connect c##dbadmin/Password1;
```

```
ALTER USER c##StudentB_1701497
  DEFAULT ROLE
  c##labclassB;
```

--3

```
GRANT CREATE ANY Procedure TO c##labclassB;
```

```
/*Located in the DBA_ROLE_PRIVS view - DBA view lists roles
granted to users and roles.
USER view lists roles granted to the current user.*/
```

-- 4

```
Connect c##dbadmin/Password1;
```

```
CREATE ROLE c##Labrolenested;
```

```
CREATE TABLE Songs(
  Track_num integer,
  Name_song varchar(25),
  Cost_song float);
```

```
insert
into Songs Values (1, 'Chronic Law - Venom', 600);
```

```
GRANT SELECT ON Songs TO c##Labrolenested;
```

```
--grant role to labclassB
GRANT c##Labrolenested TO c##labclassB;

--5
Connect c##StudentB_1701497/pass1234;

SELECT * FROM c##dbadmin.Songs;
/*Resolves becuase the role was not set which is identified by a
password
in the question*/

--Set the role for studentB
SET ROLE c##LABCLASSB IDENTIFIED BY pass1234;

SELECT * FROM c##dbadmin.Songs;
--Successful table view
```

EXERCISE 8 - REVOKE PRIVILEGES

```
--1

Connect c##dbadmin/Password1;

REVOKE CREATE ANY TABLE FROM c##labclassA;

--2

Connect c##StudentA_1701497/pass1234;

CREATE TABLE Product(
```

```
prod_id integer,  
product_name varchar(25),  
vender varchar(25),  
description varchar(30));
```

EXERCISE 9

--1

```
Connect c##dbadmin/Password1;
```

```
DESCRIBE DBA_ROLE_PRIVS
```

```
SELECT *  
FROM DBA_ROLE_PRIVS  
WHERE GRANTEE LIKE '%STUDENTA%';
```

--2

```
DROP ROLE C##LABCLASSA;
```

--3

```
SELECT *  
FROM DBA_ROLE_PRIVS  
WHERE GRANTEE LIKE '%STUDENTA%';
```

```
/*no rows selected StudentA does not have any more granted role*/
```

Screenshots of Command ran in SQL Plus

```
SQL*Plus: Release 19.0.0.0.0 - Production on Mon Feb 21 11:34:55 2022
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle. All rights reserved.

Enter user-name: c##dbadmin
Enter password:
Last Successful login time: Mon Feb 21 2022 11:31:23 -05:00

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0

SQL> create user c##test1_1701497 identified by Password1 account lock;

User created.

SQL> connect c##test1_1701497
Enter password:
ERROR:
ORA-28000: The account is locked.

Warning: You are no longer connected to ORACLE.
SQL>
```

```
SQL> Connect c##dbadmin
Enter password:
Connected.
SQL> CREATE USER c##test2_1701497 identified by Password1 PASSWORD EXPIRE;

User created.

SQL> Connect c##test2_1701497
Enter password:
ERROR:
ORA-01045: user C##TEST2_1701497 lacks CREATE SESSION privilege; logon denied

Warning: You are no longer connected to ORACLE.
SQL>
```

```
SQL> connect C##dbadmin
Enter password:
Connected.
SQL> grant create session to c##test2_1701497;

Grant succeeded.

SQL> Connect C##test2_1701497
Enter password:
ERROR:
ORA-28001: the password has expired

Changing password for C##test2_1701497
New password:
Retype new password:
Password changed
Connected.
SQL>
```

```
Connected.
SQL> grant create session to c##test2_1701497;

Grant succeeded.

SQL> Connect C##test2_1701497
Enter password:
ERROR:
ORA-28001: the password has expired

Changing password for C##test2_1701497
New password:
Retype new password:
Password changed
Connected.
SQL> Connect C##dbadmin
Enter password:
Connected.
SQL> Alter user c##test2_1701497 identified by newpassword123 password expire;

User altered.

SQL> connect c##test2_1701497/Password1
ERROR:
ORA-01017: invalid username/password; logon denied

Warning: You are no longer connected to ORACLE.
SQL> connect c##test2_1701497/newpassword123
ERROR:
ORA-28001: the password has expired

Changing password for c##test2_1701497
New password:
```

```

SQL Plus
ORA-01045: user C##TEST2_1701497 lacks CREATE SESSION privilege; logon denied

Warning: You are no longer connected to ORACLE.
SQL> connect c##test_1701497
Enter password:
ERROR:
ORA-01017: invalid username/password; logon denied

SQL> connect c##test2_1701497
Enter password:
ERROR:
ORA-01045: user C##TEST2_1701497 lacks CREATE SESSION privilege; logon denied

SQL> connect C##dbadmin
Enter password:
Connected.
SQL> grant create session to c##test2_1701497;

Grant succeeded.

SQL> Connect C##test2_1701497
Enter password:
ERROR:
ORA-28001: the password has expired

Changing password for C##test2_1701497
New password:
Retype new password:
Password changed
Connected.
SQL> Connect C##dbadmin
Enter password:
Connected.
SQL> Alter user c##test2_1701497 identified by newpassword123 password expire;

User altered.

SQL> connect c##test2_1701497/Password1
ERROR:
ORA-01017: invalid username/password; logon denied

Warning: You are no longer connected to ORACLE.
SQL> connect c##test2_1701497/newpassword123
ERROR:
ORA-28001: the password has expired

Changing password for c##test2_1701497
New password:
Retype new password:
Password changed
Connected.
SQL>

```

```

SQL> Connect C##dbadmin
Enter password:
Connected.
SQL> Alter user c##test2_1701497 identified by newpassword123 password expire;

User altered.

SQL> connect c##test2_1701497/Password1
ERROR:
ORA-01017: invalid username/password; logon denied

Warning: You are no longer connected to ORACLE.
SQL> connect c##test2_1701497/newpassword123
ERROR:
ORA-28001: the password has expired

Changing password for c##test2_1701497
New password:
Retype new password:
Password changed
Connected.
SQL> Connect c##test2_1701497/Password1
Connected.
SQL>

```

```
Warning: You are no longer connected to ORACLE.
SQL> connect c##test2_1701497/newpassword123
ERROR:
ORA-28001: the password has expired

Changing password for c##test2_1701497
New password:
Retype new password:
Password changed
Connected.
SQL> Connect c##test2_1701497/Password1
Connected.
SQL> Connect c##dbadmin/Password1
Connected.
SQL> Alter USER c##test1_1701497 account UNLOCK;

User altered.

SQL> Alter USER c##test1_1701497 password EXPIRE;

User altered.

SQL> connect c##test1_1701497/Password1
ERROR:
ORA-01045: user C##TEST1_1701497 lacks CREATE SESSION privilege; logon denied

Warning: You are no longer connected to ORACLE.
SQL> connect c##dbadmin/Password1
Connected.
SQL> Grant create session to c##test1_1701497;

Grant succeeded.

SQL> connect c##test1_1701497/Password1
ERROR:
ORA-28001: the password has expired

Changing password for c##test1_1701497
New password:
Retype new password:
Password changed
Connected.
SQL> Connect c##test1_1701497/Password2
Connected.
SQL>
```

```
SQL Plus
SQL> connect c##dbadmin/Password1;
Connected.
SQL> drop user C##test1_1701497
2 ;
drop user C##test1_1701497
*
ERROR at line 1:
ORA-01918: user 'C##TEST1_1701497' does not exist

SQL>
```

```
SQL Plus
SQL> connect c##dbadmin/Password1;
Connected.
SQL> drop user C##test1_1701497
2 ;
drop user C##test1_1701497
*
ERROR at line 1:
ORA-01918: user 'C##TEST1_1701497' does not exist

SQL> commit;

Commit complete.

SQL> Select * from dba_users where username = 'c##test1_1701497';

no rows selected

SQL>
```



```
SQL Plus
Enter password:

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0

SQL> Connect C##student_1701497/philbo$$dba;
Connected.
SQL> Create table Product(
  2   ID INTEGER,
  3   Name_Product VARCHAR(25),
  4   Price FLOAT,
  5   Description_product VARCHAR(30),
  6   ExpiryDate DATE
  7
SQL> );
SP2-0042: unknown command ")" - rest of line ignored.
SQL> Create table Product(
  2   ID INTEGER,
  3   Name_Product VARCHAR(25),
  4   Price FLOAT,
  5   Description_product VARCHAR(30),
  6   ExpiryDate DATE);
*
ERROR at line 1:
ORA-01031: insufficient privileges

SQL>
```

```
SQL> connect c##dbadmin/Password1
Connected.
SQL> GRANT CREATE TABLE TO C##student_1701497;

Grant succeeded.

SQL> Connect C##student_1701497/philbo$$dba;
Connected.
SQL>
```

```
SQL> Connect C##student_1701497/philbo$$dba;
Connected.
SQL> Create table Product(
  2   ID INTEGER,
  3   Name_Product VARCHAR(25),
  4   Price FLOAT,
  5   Description_product VARCHAR(30),
  6   ExpiryDate DATE);

Table created.

SQL>
```

```
SQL Plus
SQL> );
SP2-0042: unknown command ")" - rest of line ignored.
SQL> Create table Product(
  2   ID INTEGER,
  3   Name_Product VARCHAR(25),
  4   Price FLOAT,
  5   Description_product VARCHAR(30),
  6   ExpiryDate DATE);
*
Create table Product(
ERROR at line 1:
ORA-01031: insufficient privileges

SQL> connect c##dbadmin/Password1
Connected.
SQL> GRANT CREATE TABLE TO C##student_1701497;

Grant succeeded.

SQL> Connect C##student_1701497/philbo$$dba;
Connected.
SQL> Create table Product(
  2   ID INTEGER,
  3   Name_Product VARCHAR(25),
  4   Price FLOAT,
  5   Description_product VARCHAR(30),
  6   ExpiryDate DATE);

Table created.

SQL> insert all
  2   into Product VALUES (1001,'Jimmy Choo',500.00,'It is very nice',TO_DATE(21-02-2023,'dd-mm-yyyy'))
  3   into Product VALUES (1002,'Tom Ford',300.00,'It is goldly',TO_DATE(16-02-2023,'dd-mm-yyyy'))
  4   into Product VALUES (1003,'Victoria',300.00,'It is very good',TO_DATE(21-02-2023,'dd-mm-yyyy'))
  5   SELECT * FROM dual;
into Product VALUES (1001,'Jimmy Choo',500.00,'It is very nice',TO_DATE(21-02-2023,'dd-mm-yyyy'))
*
ERROR at line 2:
ORA-01950: no privileges on tablespace 'USERS'

SQL> insert all
  2   into Product VALUES (1001,'Jimmy Choo',500.00,'It is very nice',TO_DATE(21-02-2023,'dd-mm-yyyy'))
  3   into Product VALUES (1002,'Tom Ford',300.00,'It is goldly',TO_DATE(16-02-2023,'dd-mm-yyyy'))
  4   into Product VALUES (1003,'Victoria',300.00,'It is very good',TO_DATE(21-02-2023,'dd-mm-yyyy'))
  5
SQL> Insert into Product VALUES (1001,'Jimmy Choo',500.00,'It is very nice',TO_DATE(21-02-2023,'dd-mm-yyyy'));
Insert into Product VALUES (1001,'Jimmy Choo',500.00,'It is very nice',TO_DATE(21-02-2023,'dd-mm-yyyy'))
*
ERROR at line 1:
ORA-01950: no privileges on tablespace 'USERS'

SQL>
```

```
SQL Plus
6  ExpiryDate DATE);
Create table Product(
*
ERROR at line 1:
ORA-01031: insufficient privileges

SQL> connect c##dbadmin/Password1
Connected.
SQL> GRANT CREATE TABLE TO C##student_1701497;

Grant succeeded.

SQL> Connect C##student_1701497/philbo$$dba;
Connected.
SQL> Create table Product(
2  ID INTEGER,
3  Name_Product VARCHAR(25),
4  Price FLOAT,
5  Description_product VARCHAR(30),
6  ExpiryDate DATE);

Table created.

SQL> insert all
2  into Product VALUES (1001,'Jimmy Choo',500.00,'It is very nice',TO_DATE(21-02-2023,'dd-mm-yyyy'))
3  into Product VALUES (1002,'Tom Ford',300.00,'It is goldly',TO_DATE(16-02-2023,'dd-mm-yyyy'))
4  into Product VALUES (1003,'Victoria',300.00,'It is very good',TO_DATE(21-02-2023,'dd-mm-yyyy'))
5  SELECT * FROM dual;
into Product VALUES (1001,'Jimmy Choo',500.00,'It is very nice',TO_DATE(21-02-2023,'dd-mm-yyyy'))
*
ERROR at line 2:
ORA-01950: no privileges on tablespace 'USERS'

SQL> insert all
2  into Product VALUES (1001,'Jimmy Choo',500.00,'It is very nice',TO_DATE(21-02-2023,'dd-mm-yyyy'))
3  into Product VALUES (1002,'Tom Ford',300.00,'It is goldly',TO_DATE(16-02-2023,'dd-mm-yyyy'))
4  into Product VALUES (1003,'Victoria',300.00,'It is very good',TO_DATE(21-02-2023,'dd-mm-yyyy'))
5
SQL> Insert into Product VALUES (1001,'Jimmy Choo',500.00,'It is very nice',TO_DATE(21-02-2023,'dd-mm-yyyy'));
Insert into Product VALUES (1001,'Jimmy Choo',500.00,'It is very nice',TO_DATE(21-02-2023,'dd-mm-yyyy'))
*
ERROR at line 1:
ORA-01950: no privileges on tablespace 'USERS'

SQL> Connect sys as sysdba
Enter password:
Connected.
SQL> ALTER USER C##student_1701497 QUOTA UNLIMITED ON USERS;

User altered.

SQL>
```

```

SQL Plus
ORA-01950: no privileges on tablespace 'USERS'

SQL> insert all
  2  into Product VALUES (1001,'Jimmy Choo',500.00,'It is very nice',TO_DATE(21-02-2023,'dd-mm-yyyy'))
  3  into Product VALUES (1002,'Tom Ford',300.00,'It is goldly',TO_DATE(16-02-2023,'dd-mm-yyyy'))
  4  into Product VALUES (1003,'Victoria',300.00,'It is very good',TO_DATE(21-02-2023,'dd-mm-yyyy'))
  5
SQL> Insert into Product VALUES (1001,'Jimmy Choo',500.00,'It is very nice',TO_DATE(21-02-2023,'dd-mm-yyyy'));
Insert into Product VALUES (1001,'Jimmy Choo',500.00,'It is very nice',TO_DATE(21-02-2023,'dd-mm-yyyy'))
*
ERROR at line 1:
ORA-01950: no privileges on tablespace 'USERS'

SQL> Connect sys as sysdba
Enter password:
Connected.
SQL> ALTER USER C##student_1701497 QUOTA UNLIMITED ON USERS;

User altered.

SQL> Connect C##student_1701497/philbo$$dba;
Connected.
SQL> insert all
  2  into Product VALUES (1001,'Jimmy Choo',500.00,'It is very nice',TO_DATE(21-02-2023,'dd-mm-yyyy'))
  3  into Product VALUES (1002,'Tom Ford',300.00,'It is goldly',TO_DATE(16-02-2023,'dd-mm-yyyy'))
  4  into Product VALUES (1003,'Victoria',300.00,'It is very good',TO_DATE(21-02-2023,'dd-mm-yyyy'))
  5  SELECT * FROM dual;
  into Product VALUES (1001,'Jimmy Choo',500.00,'It is very nice',TO_DATE(21-02-2023,'dd-mm-yyyy'))
*
ERROR at line 2:
ORA-01847: day of month must be between 1 and last day of month

SQL> insert all
  2  into Product VALUES (1001,'Jimmy Choo',500.00,'It is very nice',TO_DATE(21/02/2023,'dd/mm/yyyy'))
  3  into Product VALUES (1002,'Tom Ford',300.00,'It is goldly',TO_DATE(16/04/2023,'dd/mm/yyyy'))
  4  into Product VALUES (1003,'Victoria',300.00,'It is very good',TO_DATE(21/02/2024,'dd/mm/yyyy'))
  5  SELECT * FROM dual;
  into Product VALUES (1001,'Jimmy Choo',500.00,'It is very nice',TO_DATE(21/02/2023,'dd/mm/yyyy'))
*
ERROR at line 2:
ORA-01858: a non-numeric character was found where a numeric was expected

SQL> insert all
  2  into Product VALUES (1001,'Jimmy Choo',500.00,'It is very nice',TO_DATE('21/02/2023','dd/mm/yyyy'))
  3  into Product VALUES (1002,'Tom Ford',300.00,'It is goldly',TO_DATE('16/04/2023','dd/mm/yyyy'))
  4  into Product VALUES (1003,'Victoria',300.00,'It is very good',TO_DATE('21/02/2024','dd/mm/yyyy'))
  5  SELECT * FROM dual;

3 rows created.

SQL>

```

```

SQL Plus
3 rows created.

SQL> Connect c##test2_1701497/Password2
ERROR:
ORA-01017: invalid username/password; logon denied

Warning: You are no longer connected to ORACLE.
SQL> Connect c##test2_1701497/Password1
Connected.
SQL> UPDATE Product
  2  SET Price = 950.00
  3  WHERE ID = 101;
UPDATE Product
*
ERROR at line 1:
ORA-00942: table or view does not exist

SQL>

```

```
SQL Plus
UPDATE Product
*
ERROR at line 1:
ORA-00942: table or view does not exist

SQL> UPDATE Product
2 SET Price = 950.00
3 WHERE ID = 1001;
UPDATE Product
*
ERROR at line 1:
ORA-00942: table or view does not exist

SQL> DELETE FROM Product
2 WHERE ID = 1003;
DELETE FROM Product
*
ERROR at line 1:
ORA-00942: table or view does not exist

SQL>
```

```
SQL Plus
5 SELECT * FROM dual;
into Product VALUES (1001,'Jimmy Choo',500.00,'It is very nice',TO_DATE(21/02/2023,'dd/mm/yyyy'))
*
ERROR at line 2:
ORA-01858: a non-numeric character was found where a numeric was expected

SQL> insert all
2 into Product VALUES (1001,'Jimmy Choo',500.00,'It is very nice',TO_DATE('21/02/2023','dd/mm/yyyy'))
3 into Product VALUES (1002,'Tom Ford',300.00,'It is goldly',TO_DATE('16/04/2023','dd/mm/yyyy'))
4 into Product VALUES (1003,'Victoria',300.00,'It is very good',TO_DATE('21/02/2024','dd/mm/yyyy'))
5 SELECT * FROM dual;

3 rows created.

SQL> Connect c##test2_1701497/Password2
ERROR:
ORA-01017: invalid username/password; logon denied

Warning: You are no longer connected to ORACLE.
SQL> Connect c##test2_1701497/Password1
Connected.
SQL> UPDATE Product
2 SET Price = 950.00
3 WHERE ID = 101;
UPDATE Product
*
ERROR at line 1:
ORA-00942: table or view does not exist

SQL> UPDATE Product
2 SET Price = 950.00
3 WHERE ID = 1001;
UPDATE Product
*
ERROR at line 1:
ORA-00942: table or view does not exist

SQL> DELETE FROM Product
2 WHERE ID = 1003;
DELETE FROM Product
*
ERROR at line 1:
ORA-00942: table or view does not exist

SQL> INSERT INTO Product VALUES (1004,'Supligen',250.00,'Get some energy',TO_DATE('21/02/2025','dd/mm/yyyy'));
INSERT INTO Product VALUES (1004,'Supligen',250.00,'Get some energy',TO_DATE('21/02/2025','dd/mm/yyyy'))
*
ERROR at line 1:
ORA-00942: table or view does not exist

SQL>
```

```
SQL Plus
ORA-01858: a non-numeric character was found where a numeric was expected

SQL> insert all
2  into Product VALUES (1001,'Jimmy Choo',500.00,'It is very nice',TO_DATE('21/02/2023','dd/mm/yyyy'))
3  into Product VALUES (1002,'Tom Ford',300.00,'It is goldly',TO_DATE('16/04/2023','dd/mm/yyyy'))
4  into Product VALUES (1003,'Victoria',300.00,'It is very good',TO_DATE('21/02/2024','dd/mm/yyyy'))
5  SELECT * FROM dual;

3 rows created.

SQL> Connect c##test2_1701497/Password2
ERROR:
ORA-01017: invalid username/password; logon denied

Warning: You are no longer connected to ORACLE.
SQL> Connect c##test2_1701497/Password1
Connected.
SQL> UPDATE Product
2  SET Price = 950.00
3  WHERE ID = 101;
UPDATE Product
*
ERROR at line 1:
ORA-00942: table or view does not exist

SQL> UPDATE Product
2  SET Price = 950.00
3  WHERE ID = 1001;
UPDATE Product
*
ERROR at line 1:
ORA-00942: table or view does not exist

SQL> DELETE FROM Product
2  WHERE ID = 1003;
DELETE FROM Product
*
ERROR at line 1:
ORA-00942: table or view does not exist

SQL> INSERT INTO Product VALUES (1004,'Supligen',250.00,'Get some energy',TO_DATE('21/02/2025','dd/mm/yyyy'));
INSERT INTO Product VALUES (1004,'Supligen',250.00,'Get some energy',TO_DATE('21/02/2025','dd/mm/yyyy'))
*
ERROR at line 1:
ORA-00942: table or view does not exist

SQL> Select * from Product;
Select * from Product
*
ERROR at line 1:
ORA-00942: table or view does not exist

SQL>
```

```
SQL Plus
ERROR:
ORA-01017: invalid username/password; logon denied

Warning: You are no longer connected to ORACLE.
SQL> Connect c##test2_1701497/Password1
Connected.
SQL> UPDATE Product
  2   SET Price = 950.00
  3   WHERE ID = 101;
UPDATE Product
*
ERROR at line 1:
ORA-00942: table or view does not exist

SQL> UPDATE Product
  2   SET Price = 950.00
  3   WHERE ID = 1001;
UPDATE Product
*
ERROR at line 1:
ORA-00942: table or view does not exist

SQL> DELETE FROM Product
  2   WHERE ID = 1003;
DELETE FROM Product
*
ERROR at line 1:
ORA-00942: table or view does not exist

SQL> INSERT INTO Product VALUES (1004,'Supligen',250.00,'Get some energy',TO_DATE('21/02/2025','dd/mm/yyyy'));
INSERT INTO Product VALUES (1004,'Supligen',250.00,'Get some energy',TO_DATE('21/02/2025','dd/mm/yyyy'))
*
ERROR at line 1:
ORA-00942: table or view does not exist

SQL> Select * from Product;
Select * from Product
*
ERROR at line 1:
ORA-00942: table or view does not exist

SQL> CONNECT c##Student_1701497/philbo$$dba;
Connected.
SQL> GRANT UPDATE, DELETE, INSERT, SELECT ON Product TO c##test2_1701497;

Grant succeeded.

SQL>
```

```
SQL Plus
ERROR at line 1:
ORA-00942: table or view does not exist

SQL> Connect c##test2_1701497/Password1;
Connected.
SQL> GRANT SELECT , INSERT , UPDATE , DELETE ON Product TO c##test2_1701497;
GRANT SELECT , INSERT , UPDATE , DELETE ON Product TO c##test2_1701497
*
ERROR at line 1:
ORA-00942: table or view does not exist

SQL> SHOW TABLES
SP2-0158: unknown SHOW option "TABLES"
SQL> CONNECT c##Student_1701497/philbo$$dba;
Connected.
SQL> RANT SELECT , INSERT , UPDATE , DELETE ON Product TO c##test2_1701497;
SP2-0734: unknown command beginning "RANT SELEC..." - rest of line ignored.
SQL> CONNECT c##Student_1701497/philbo$$dba;
Connected.
SQL> GRANT SELECT , INSERT , UPDATE , DELETE ON Product TO c##test2_1701497;

Grant succeeded.

SQL> Connect c##test2_1701497/Password1;
Connected.
SQL> UPDATE c##Student_1701497.Product
  2   SET Price = 950.00
  3   WHERE ID = 1001;

1 row updated.

SQL> DELETE FROM c##Student_1701497.Product
  2   WHERE ID = 1003;

1 row deleted.

SQL> INSERT INTO c##Student_1701497.Product VALUES (1004,'Supligen',250.00,'Get some energy',TO_DATE('21/02/2025','dd/mm
/yyyy'));

1 row created.

SQL> Select * from c##Student_1701497.Product;

      ID NAME_PRODUCT          PRICE DESCRIPTION_PRODUCT
-----
EXPIRYDAT
-----
    1001 Jimmy Choo             950 It is very nice
21-FEB-23
    1002 Tom Ford               300 It is goldly
16-APR-23
    1004 Supligen              250 Get some energy
21-FEB-25

SQL>
```



```
SQL Plus
SQL> Connect c##dbadmin/Password1;
Connected.
SQL> CREATE USER c##dummy_1701497 IDENTIFIED BY Password1;

User created.

SQL> GRANT CREATE TABLE TO c##dummy_1701497;

Grant succeeded.

SQL> GRANT ALL TO c##dummy_1701497;
GRANT ALL TO c##dummy_1701497
*
ERROR at line 1:
ORA-01919: role 'ALL' does not exist

SQL>
```

```
SQL Plus
SQL> Connect c##dummy_1701497/Password1;
ERROR:
ORA-01045: user C##DUMMY_1701497 lacks CREATE SESSION privilege; logon denied

Warning: You are no longer connected to ORACLE.
SQL> Connect c##dbadmin/Password1;
Connected.
SQL> Grant create session to c##dummy_1701497;

Grant succeeded.

SQL>
```

```

SQL Plus
Warning: You are no longer connected to ORACLE.
SQL> Connect c##dbadmin/Password1;
Connected.
SQL> Grant create session to c##dummy_1701497;

Grant succeeded.

SQL> Connect c##dummy_1701497/Password1;
Connected.
SQL> Create table Product(
  2   ID INTEGER,
  3   Name_Product VARCHAR(25),
  4   Price FLOAT,
  5   Description_product VARCHAR(30),
  6   ExpiryDate DATE);

Table created.

SQL> DROP TABLE Product;

Table dropped.

SQL> Create table Dummy(
  2   ID INTEGER,
  3   Name_Product VARCHAR(25),
  4   Price FLOAT,
  5   Description_product VARCHAR(30),
  6   ExpiryDate DATE);

Table created.

SQL> Connect c##dbadmin/Password1;
Connected.
SQL> DROP USER c##dummy_1701497;
DROP USER c##dummy_1701497
*
ERROR at line 1:
ORA-01922: CASCADE must be specified to drop 'C##DUMMY_1701497'

SQL>

```

```

SQL Plus
Warning: You are no longer connected to ORACLE.
SQL> Connect c##dbadmin/Password1;
Connected.
SQL> Grant create session to c##dummy_1701497;

Grant succeeded.

SQL> Connect c##dummy_1701497/Password1;
Connected.
SQL> Create table Product(
  2   ID INTEGER,
  3   Name_Product VARCHAR(25),
  4   Price FLOAT,
  5   Description_product VARCHAR(30),
  6   ExpiryDate DATE);

Table created.

SQL> DROP TABLE Product;

Table dropped.

SQL> Create table Dummy(
  2   ID INTEGER,
  3   Name_Product VARCHAR(25),
  4   Price FLOAT,
  5   Description_product VARCHAR(30),
  6   ExpiryDate DATE);

Table created.

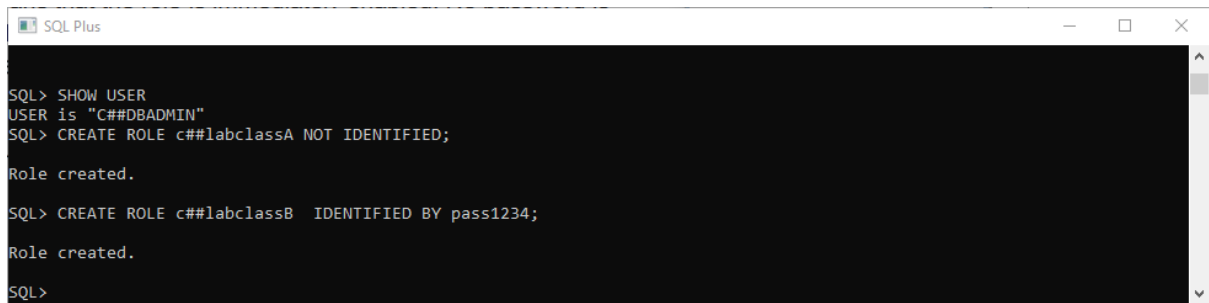
SQL> Connect c##dbadmin/Password1;
Connected.
SQL> DROP USER c##dummy_1701497;
DROP USER c##dummy_1701497
*
ERROR at line 1:
ORA-01922: CASCADE must be specified to drop 'C##DUMMY_1701497'

SQL> DROP USER c##dummy_1701497 CASCADE;

User dropped.

SQL>

```



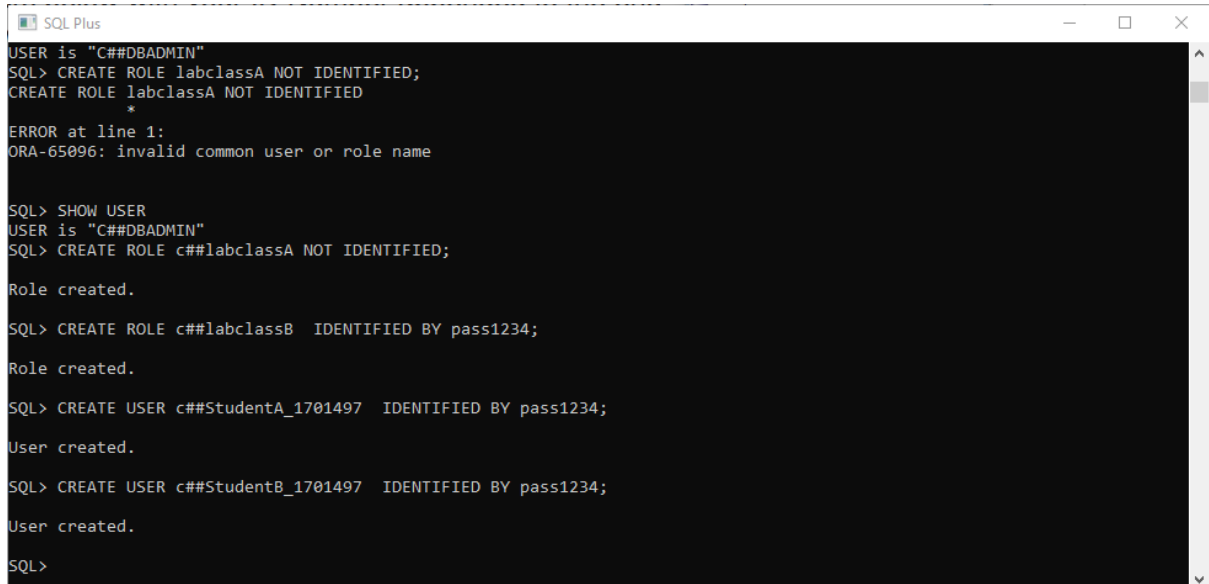
```
SQL Plus
SQL> SHOW USER
USER is "C##DBADMIN"
SQL> CREATE ROLE c##labclassA NOT IDENTIFIED;

Role created.

SQL> CREATE ROLE c##labclassB IDENTIFIED BY pass1234;

Role created.

SQL>
```



```
SQL Plus
USER is "C##DBADMIN"
SQL> CREATE ROLE labclassA NOT IDENTIFIED;
CREATE ROLE labclassA NOT IDENTIFIED
*
ERROR at line 1:
ORA-65096: invalid common user or role name

SQL> SHOW USER
USER is "C##DBADMIN"
SQL> CREATE ROLE c##labclassA NOT IDENTIFIED;

Role created.

SQL> CREATE ROLE c##labclassB IDENTIFIED BY pass1234;

Role created.

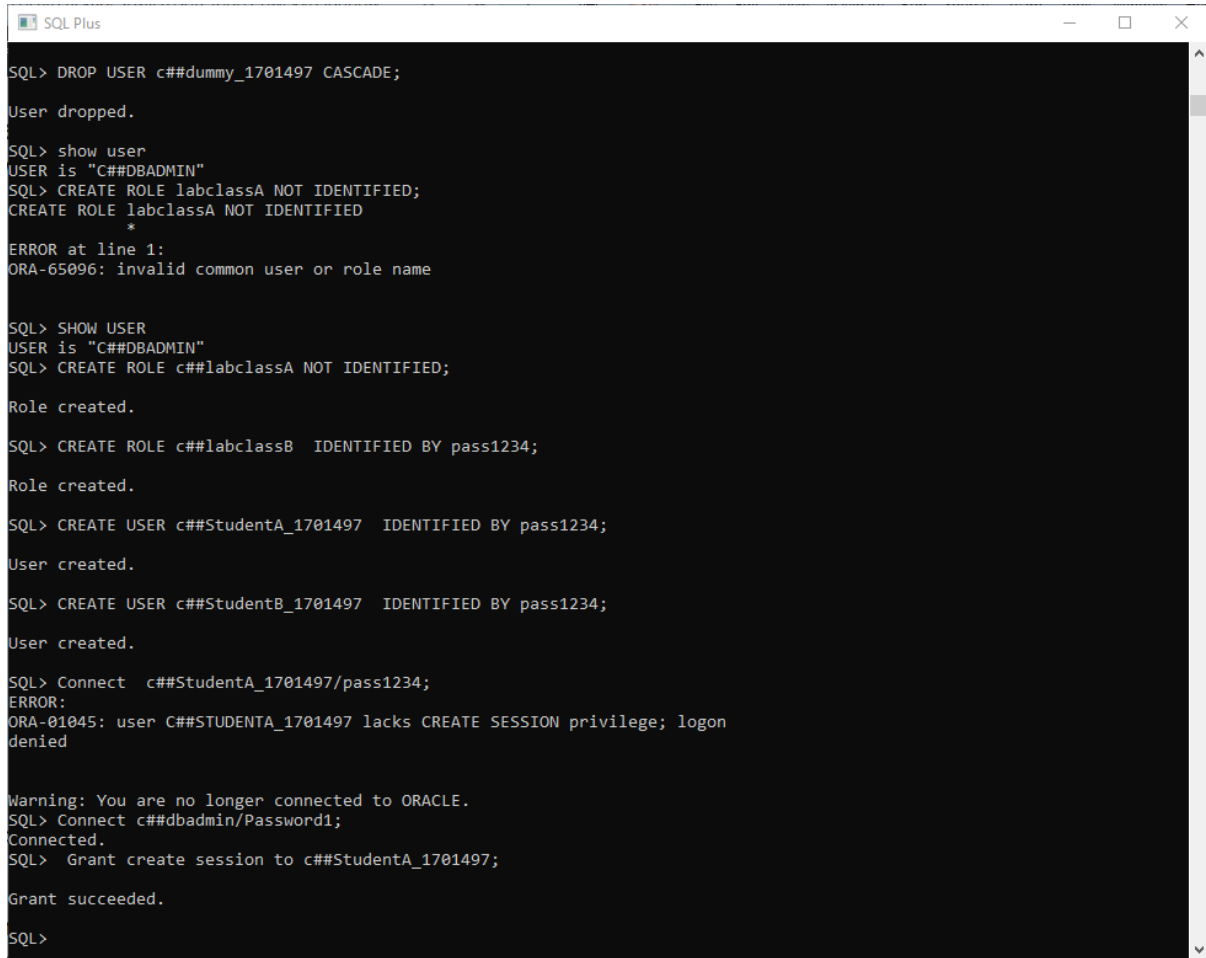
SQL> CREATE USER c##StudentA_1701497 IDENTIFIED BY pass1234;

User created.

SQL> CREATE USER c##StudentB_1701497 IDENTIFIED BY pass1234;

User created.

SQL>
```



```
SQL> DROP USER c##dummy_1701497 CASCADE;

User dropped.

SQL> show user
USER is "C##DBADMIN"
SQL> CREATE ROLE labclassA NOT IDENTIFIED;
CREATE ROLE labclassA NOT IDENTIFIED
*
ERROR at line 1:
ORA-65096: invalid common user or role name

SQL> SHOW USER
USER is "C##DBADMIN"
SQL> CREATE ROLE c##labclassA NOT IDENTIFIED;

Role created.

SQL> CREATE ROLE c##labclassB IDENTIFIED BY pass1234;

Role created.

SQL> CREATE USER c##StudentA_1701497 IDENTIFIED BY pass1234;

User created.

SQL> CREATE USER c##StudentB_1701497 IDENTIFIED BY pass1234;

User created.

SQL> Connect c##StudentA_1701497/pass1234;
ERROR:
ORA-01045: user C##STUDENTA_1701497 lacks CREATE SESSION privilege; logon
denied

Warning: You are no longer connected to ORACLE.
SQL> Connect c##dbadmin/Password1;
Connected.
SQL> Grant create session to c##StudentA_1701497;

Grant succeeded.

SQL>
```

```

SQL Plus
SQL> SHOW USER
USER is "C##DBADMIN"
SQL> CREATE ROLE c##labclassA NOT IDENTIFIED;
Role created.
SQL> CREATE ROLE c##labclassB IDENTIFIED BY pass1234;
Role created.
SQL> CREATE USER c##StudentA_1701497 IDENTIFIED BY pass1234;
User created.
SQL> CREATE USER c##StudentB_1701497 IDENTIFIED BY pass1234;
User created.
SQL> Connect c##StudentA_1701497/pass1234;
ERROR:
ORA-01045: user C##STUDENTA_1701497 lacks CREATE SESSION privilege; logon
denied
Warning: You are no longer connected to ORACLE.
SQL> Connect c##dbadmin/Password1;
Connected.
SQL> Grant create session to c##StudentA_1701497;
Grant succeeded.
SQL> Connect c##StudentA_1701497/pass1234;
Connected.
SQL> Create table Sales(
  2     Sales_Id VARCHAR(25),
  3     Product VARCHAR(25),
  4     Unitcost FLOAT,
  5     quantity INTEGER);
Create table Sales(
*
ERROR at line 1:
ORA-01031: insufficient privileges
SQL>

```

```

SQL Plus
User created.
SQL> Connect c##StudentA_1701497/pass1234;
ERROR:
ORA-01045: user C##STUDENTA_1701497 lacks CREATE SESSION privilege; logon
denied
Warning: You are no longer connected to ORACLE.
SQL> Connect c##dbadmin/Password1;
Connected.
SQL> Grant create session to c##StudentA_1701497;
Grant succeeded.
SQL> Connect c##StudentA_1701497/pass1234;
Connected.
SQL> Create table Sales(
  2     Sales_Id VARCHAR(25),
  3     Product VARCHAR(25),
  4     Unitcost FLOAT,
  5     quantity INTEGER);
Create table Sales(
*
ERROR at line 1:
ORA-01031: insufficient privileges
SQL> Connect c##dbadmin/Password1;
Connected.
SQL> GRANT c##labclassA TO c##StudentA_1701497;
Grant succeeded.
SQL>

```

```
SQL Plus
Warning: You are no longer connected to ORACLE.
SQL> Connect c##dbadmin/Password1;
Connected.
SQL> Grant create session to c##StudentA_1701497;

Grant succeeded.

SQL> Connect c##StudentA_1701497/pass1234;
Connected.
SQL> Create table Sales(
  2     Sales_Id VARCHAR(25),
  3     Product VARCHAR(25),
  4     Unitcost FLOAT,
  5     quantity INTEGER);
Create table Sales(
*
ERROR at line 1:
ORA-01031: insufficient privileges

SQL> Connect c##dbadmin/Password1;
Connected.
SQL> GRANT c##labclassA TO c##StudentA_1701497;

Grant succeeded.

SQL> ALTER USER c##StudentA_1701497
  2     DEFAULT ROLE
  3     c##labclassA;

User altered.

SQL>
```

```
SQL Plus
User created.

SQL> Connect c##StudentA_1701497/pass1234;
ERROR:
ORA-01045: user C##STUDENTA_1701497 lacks CREATE SESSION privilege; logon
denied

Warning: You are no longer connected to ORACLE.
SQL> Connect c##dbadmin/Password1;
Connected.
SQL> Grant create session to c##StudentA_1701497;

Grant succeeded.

SQL> Connect c##StudentA_1701497/pass1234;
Connected.
SQL> Create table Sales(
  2     Sales_Id VARCHAR(25),
  3     Product VARCHAR(25),
  4     Unitcost FLOAT,
  5     quantity INTEGER);
Create table Sales(
*
ERROR at line 1:
ORA-01031: insufficient privileges

SQL> Connect c##dbadmin/Password1;
Connected.
SQL> GRANT c##labclassA TO c##StudentA_1701497;

Grant succeeded.

SQL> ALTER USER c##StudentA_1701497
  2     DEFAULT ROLE
  3     c##labclassA;

User altered.

SQL> GRANT CREATE ANY TABLE TO c##labclassA;
```

```
SQL Plus

SQL> SHOW USER
USER is "C##DBADMIN"
SQL> CREATE ROLE c##labclassA NOT IDENTIFIED;

Role created.

SQL> CREATE ROLE c##labclassB IDENTIFIED BY pass1234;

Role created.

SQL> CREATE USER c##StudentA_1701497 IDENTIFIED BY pass1234;

User created.

SQL> CREATE USER c##StudentB_1701497 IDENTIFIED BY pass1234;

User created.

SQL> Connect c##StudentA_1701497/pass1234;
ERROR:
ORA-01045: user C##STUDENTA_1701497 lacks CREATE SESSION privilege; logon
denied

Warning: You are no longer connected to ORACLE.
SQL> Connect c##dbadmin/Password1;
Connected.
SQL> Grant create session to c##StudentA_1701497;

Grant succeeded.

SQL> Connect c##StudentA_1701497/pass1234;
Connected.
SQL> Create table Sales(
  2     Sales_Id VARCHAR(25),
  3     Product VARCHAR(25),
  4     Unitcost FLOAT,
  5     quantity INTEGER);
Create table Sales(
*
ERROR at line 1:
ORA-01031: insufficient privileges

SQL> Connect c##dbadmin/Password1;
Connected.
SQL> GRANT c##labclassA TO c##StudentA_1701497;

Grant succeeded.

SQL> ALTER USER c##StudentA_1701497
  2     DEFAULT ROLE
  3     c##labclassA;

User altered.

SQL> GRANT CREATE ANY TABLE TO c##labclassA;

Grant succeeded.

SQL> ;
```

```
SQL Plus

Grant succeeded.

SQL> Connect c##StudentA_1701497/pass1234;
Connected.
SQL> Create table Sales(
  2     Sales_Id VARCHAR(25),
  3     Product VARCHAR(25),
  4     Unitcost FLOAT,
  5     quantity INTEGER);
Create table Sales(
*
ERROR at line 1:
ORA-01031: insufficient privileges

SQL> Connect c##dbadmin/Password1;
Connected.
SQL> GRANT c##labclassA TO c##StudentA_1701497;

Grant succeeded.

SQL> ALTER USER c##StudentA_1701497
  2     DEFAULT ROLE
  3     c##labclassA;

User altered.

SQL> GRANT CREATE ANY TABLE TO c##labclassA;

Grant succeeded.

SQL> ;Connect c##StudentA_1701497/pass1234;
SP2-0734: unknown command beginning ";Connect ..." - rest of line ignored.
SQL> Connect c##StudentA_1701497/pass1234;
Connected.
SQL> Create table Sales(
  2     Sales_Id VARCHAR(25),
  3     Product VARCHAR(25),
  4     Unitcost FLOAT,
  5     quantity INTEGER);

Table created.

SQL>
```



```
SQL Plus
SQL> ;Connect c##StudentA_1701497/pass1234;
SP2-0734: unknown command beginning ";Connect ..." - rest of line ignored.
SQL> Connect c##StudentA_1701497/pass1234;
Connected.
SQL> Create table Sales(
  2   Sales_Id VARCHAR(25),
  3   Product VARCHAR(25),
  4   Unitcost FLOAT,
  5   quantity INTEGER);

Table created.

SQL>      insert all
  2      into Sales VALUES (A001,'Apple',20.00,10)
  3      into Sales VALUES (B002,'Mango',40.00,70)
  4      into Sales VALUES (C003,'Pear',900.00,20)
  5      SELECT * FROM dual;
      into Sales VALUES (C003,'Pear',900.00,20)
      *
ERROR at line 4:
ORA-00904: "C003": invalid identifier

SQL>      insert all
  2      into Sales VALUES ('A001','Apple',20.00,10)
  3      into Sales VALUES ('B002','Mango',40.00,70)
  4      into Sales VALUES ('C003','Pear',900.00,20)
  5      SELECT * FROM dual;
      into Sales VALUES ('A001','Apple',20.00,10)
      *
ERROR at line 2:
ORA-01950: no privileges on tablespace 'USERS'

SQL>
```

```
SQL Plus
SQL> Connect c##StudentA_1701497/pass1234;
Connected.
SQL> Create table Sales(
  2   Sales_Id VARCHAR(25),
  3   Product VARCHAR(25),
  4   Unitcost FLOAT,
  5   quantity INTEGER);

Table created.

SQL>      insert all
  2      into Sales VALUES (A001,'Apple',20.00,10)
  3      into Sales VALUES (B002,'Mango',40.00,70)
  4      into Sales VALUES (C003,'Pear',900.00,20)
  5      SELECT * FROM dual;
      into Sales VALUES (C003,'Pear',900.00,20)
      *
ERROR at line 4:
ORA-00904: "C003": invalid identifier

SQL>      insert all
  2      into Sales VALUES ('A001','Apple',20.00,10)
  3      into Sales VALUES ('B002','Mango',40.00,70)
  4      into Sales VALUES ('C003','Pear',900.00,20)
  5      SELECT * FROM dual;
      into Sales VALUES ('A001','Apple',20.00,10)
      *
ERROR at line 2:
ORA-01950: no privileges on tablespace 'USERS'

SQL>      Connect c##dbadmin/Password1;
Connected.
SQL> ALTER USER c##StudentA_1701497 QUOTA UNLIMITED ON USERS;

User altered.

SQL>
```

```

SQL Plus
SQL> insert all
2      into Sales VALUES ('A001','Apple',20.00,10)
3      into Sales VALUES ('B002','Mango',40.00,70)
4      into Sales VALUES ('C003','Pear',900.00,20)
5      SELECT * FROM dual;
      into Sales VALUES ('A001','Apple',20.00,10)
      *
ERROR at line 2:
ORA-01950: no privileges on tablespace 'USERS'

SQL> Connect c##dbadmin/Password1;
Connected.
SQL> ALTER USER c##StudentA_1701497 QUOTA UNLIMITED ON USERS;

User altered.

SQL> insert all
2      into Sales VALUES ('A001','Apple',20.00,10)
3      into Sales VALUES ('B002','Mango',40.00,70)
4      into Sales VALUES ('C003','Pear',900.00,20)
5      SELECT * FROM dual;
      into Sales VALUES ('C003','Pear',900.00,20)
      *
ERROR at line 4:
ORA-00942: table or view does not exist

SQL> Connect c##StudentA_1701497/pass1234;
Connected.
SQL> insert all
2      into Sales VALUES ('A001','Apple',20.00,10)
3      into Sales VALUES ('B002','Mango',40.00,70)
4      into Sales VALUES ('C003','Pear',900.00,20)
5      SELECT * FROM dual;

3 rows created.

SQL>

```

```

SQL Plus
ORA-01950: no privileges on tablespace 'USERS'

SQL> Connect c##dbadmin/Password1;
Connected.
SQL> ALTER USER c##StudentA_1701497 QUOTA UNLIMITED ON USERS;

User altered.

SQL> insert all
2      into Sales VALUES ('A001','Apple',20.00,10)
3      into Sales VALUES ('B002','Mango',40.00,70)
4      into Sales VALUES ('C003','Pear',900.00,20)
5      SELECT * FROM dual;
      into Sales VALUES ('C003','Pear',900.00,20)
      *
ERROR at line 4:
ORA-00942: table or view does not exist

SQL> Connect c##StudentA_1701497/pass1234;
Connected.
SQL> insert all
2      into Sales VALUES ('A001','Apple',20.00,10)
3      into Sales VALUES ('B002','Mango',40.00,70)
4      into Sales VALUES ('C003','Pear',900.00,20)
5      SELECT * FROM dual;

3 rows created.

SQL> Select * from Sales;

SALES_ID          PRODUCT          UNITCOST    QUANTITY
-----
A001              Apple              20             10
B002              Mango              40             70
C003              Pear             900             20

SQL>

```

```

SQL Plus
SQL> insert all
2      into Sales VALUES ('A001','Apple',20.00,10)
3      into Sales VALUES ('B002','Mango',40.00,70)
4      into Sales VALUES ('C003','Pear',900.00,20)
5      SELECT * FROM dual;

3 rows created.

SQL> Select * from Sales;

SALES_ID          PRODUCT                UNITCOST    QUANTITY
-----
A001              Apple                  20          10
B002              Mango                  40          70
C003              Pear                  900         20

SQL> Connect c##StudentB_1701497/pass1234;
ERROR:
ORA-01045: user C##STUDENTB_1701497 lacks CREATE SESSION privilege; logon
denied

Warning: You are no longer connected to ORACLE.
SQL> Connect c##dbadmin/Password1;
Connected.
SQL> Grant create session to c##StudentB_1701497;

Grant succeeded.

SQL> Connect c##StudentB_1701497/pass1234;
Connected.
SQL> Select * from c##StudentA_1701497.Sales;
Select * from c##StudentA_1701497.Sales
*
ERROR at line 1:
ORA-00942: table or view does not exist

SQL>

```

```

SQL Plus
SQL> Select * from Sales;

SALES_ID          PRODUCT                UNITCOST    QUANTITY
-----
A001              Apple                  20          10
B002              Mango                  40          70
C003              Pear                  900         20

SQL> Connect c##StudentB_1701497/pass1234;
ERROR:
ORA-01045: user C##STUDENTB_1701497 lacks CREATE SESSION privilege; logon
denied

Warning: You are no longer connected to ORACLE.
SQL> Connect c##dbadmin/Password1;
Connected.
SQL> Grant create session to c##StudentB_1701497;

Grant succeeded.

SQL> Connect c##StudentB_1701497/pass1234;
Connected.
SQL> Select * from c##StudentA_1701497.Sales;
Select * from c##StudentA_1701497.Sales
*
ERROR at line 1:
ORA-00942: table or view does not exist

SQL> Connect c##StudentA_1701497/pass1234;
Connected.
SQL> GRANT SELECT ON Sales TO c##StudentB_1701497
2 ;

Grant succeeded.

SQL>

```

```

SQL Plus
SQL> Connect c##StudentA_1701497/pass1234;
Connected.
SQL> insert all
  2      into Sales VALUES ('A001','Apple',20.00,10)
  3      into Sales VALUES ('B002','Mango',40.00,70)
  4      into Sales VALUES ('C003','Pear',900.00,20)
  5      SELECT * FROM dual;

3 rows created.

SQL> Select * from Sales;

SALES_ID          PRODUCT          UNITCOST    QUANTITY
-----
A001              Apple              20          10
B002              Mango              40          70
C003              Pear              900         20

SQL> Connect c##StudentB_1701497/pass1234;
ERROR:
ORA-01045: user C##STUDENTB_1701497 lacks CREATE SESSION privilege; logon
denied

Warning: You are no longer connected to ORACLE.
SQL> Connect c##dbadmin/Password1;
Connected.
SQL> Grant create session to c##StudentB_1701497;

Grant succeeded.

SQL> Connect c##StudentB_1701497/pass1234;
Connected.
SQL> Select * from c##StudentA_1701497.Sales;
Select * from c##StudentA_1701497.Sales
*
ERROR at line 1:
ORA-00942: table or view does not exist

SQL> Connect c##StudentA_1701497/pass1234;
Connected.
SQL> GRANT SELECT ON Sales TO c##StudentB_1701497
  2  ;

Grant succeeded.

SQL> Connect c##StudentB_1701497/pass1234;
Connected.
SQL> Select * from c##StudentA_1701497.Sales;

SALES_ID          PRODUCT          UNITCOST    QUANTITY
-----
A001              Apple              20          10
B002              Mango              40          70
C003              Pear              900         20

SQL>

```

```

SQL> Connect c##dbadmin/Password1;
Connected.
SQL> GRANT c##labclassB TO c##StudentB_1701497;

Grant succeeded.

SQL>

```

```

SQL Plus
DELEGATE_OPTION          VARCHAR2(3)
DEFAULT_ROLE             VARCHAR2(3)
COMMON                   VARCHAR2(3)
INHERITED                 VARCHAR2(3)

SQL> DESCRIBE DBA_ROLES
Name                      Null?    Type
-----
ROLE                      NOT NULL VARCHAR2(128)
ROLE_ID                   NOT NULL NUMBER
PASSWORD_REQUIRED         VARCHAR2(8)
AUTHENTICATION_TYPE       VARCHAR2(11)
COMMON                    VARCHAR2(3)
ORACLE_MAINTAINED         VARCHAR2(1)
INHERITED                 VARCHAR2(3)
IMPLICIT                  VARCHAR2(3)
EXTERNAL_NAME             VARCHAR2(4000)

SQL> SELECT * FROM DBA_ROLES
2      WHERE ROLE LIKE '%C##LABCLASS%'
3      ;

ROLE
-----
ROLE_ID PASSWORD AUTHENTICAT COM O INH IMP
-----
EXTERNAL_NAME
-----
C##LABCLASSA
113 NO      NONE      YES N NO  NO

C##LABCLASSB
114 YES     PASSWORD  YES N NO  NO

ROLE
-----
ROLE_ID PASSWORD AUTHENTICAT COM O INH IMP
-----
EXTERNAL_NAME
-----

SQL>

```

```

SQL> Connect c##dbadmin/Password1;
Connected.
SQL> ALTER USER c##StudentB_1701497
2      DEFAULT ROLE
3      c##labclassB;

User altered.

SQL>

```

```

SQL> GRANT CREATE ANY Procedure TO c##labclassB;

Grant succeeded.

SQL>

```

```
SQL> Connect c##dbadmin/Password1;
Connected.
SQL> ALTER USER c##StudentB_1701497
  2     DEFAULT ROLE
  3     c##labclassB;

User altered.

SQL> GRANT CREATE ANY Procedure TO c##labclassB;

Grant succeeded.

SQL> CREATE ROLE c##Labrolenested;

Role created.

SQL> CREATE TABLE Songs(
  2     Track_num integer,
  3     Name_song varchar(25),
  4     Cost_song float);

Table created.

SQL>
```

```
SQL> Connect c##dbadmin/Password1;
Connected.
SQL> ALTER USER c##StudentB_1701497
  2     DEFAULT ROLE
  3     c##labclassB;

User altered.

SQL> GRANT CREATE ANY Procedure TO c##labclassB;

Grant succeeded.

SQL> CREATE ROLE c##Labrolenested;

Role created.

SQL> CREATE TABLE Songs(
  2     Track_num integer,
  3     Name_song varchar(25),
  4     Cost_song float);

Table created.

SQL> GRANT SELECT ON Songs TO c##Labrolenested;

Grant succeeded.

SQL> GRANT c##Labrolenested TO c##labclassB;

Grant succeeded.

SQL>
```

```
SQL> Connect c##dbadmin/Password1;
Connected.
SQL> ALTER USER c##StudentB_1701497
  2     DEFAULT ROLE
  3     c##labclassB;

User altered.

SQL> GRANT CREATE ANY Procedure TO c##labclassB;

Grant succeeded.

SQL> CREATE ROLE c##Labrolenested;

Role created.

SQL> CREATE TABLE Songs(
  2     Track_num integer,
  3     Name_song varchar(25),
  4     Cost_song float);

Table created.

SQL> GRANT SELECT ON Songs TO c##Labrolenested;

Grant succeeded.

SQL> GRANT c##Labrolenested TO c##labclassB;

Grant succeeded.

SQL> insert
  2 into Songs Values (1,'Chronic Law - Venom',600);

1 row created.

SQL>
```

```
Table created.

SQL> GRANT SELECT ON Songs TO c##Labrolenested;

Grant succeeded.

SQL> GRANT c##Labrolenested TO c##labclassB;

Grant succeeded.

SQL> insert
  2  into Songs Values (1,'Chronic Law - Venom',600);

1 row created.

SQL> Connect  c##StudentB_1701497/pass1234;
Connected.
SQL> SELECT * FROM c##dbadmin.songs
  2  ;
SELECT * FROM c##dbadmin.songs
          *
ERROR at line 1:
ORA-00942: table or view does not exist
```

```
SQL> Connect c##dbadmin/Password1;
Connected.
SQL> REVOKE CREATE ANY TABLE FROM c##labclassA;

Revoke succeeded.

SQL>
```



```
SQL> Connect c##StudentA_1701497/pass1234;
Connected.
SQL> CREATE TABLE Product(
  2     prod_id integer,
  3     product_name varchar(25),
  4     vender varchar(25),
  5     description varchar(30));
CREATE TABLE Product(
*
ERROR at line 1:
ORA-01031: insufficient privileges

SQL>
```

```
C##ADMIN
DBA
NO NO YES NO NO

GRANTEE
-----
GRANTED_ROLE
-----
ADM DEL DEF COM INH
-- -- -- -- --

C##STUDENTB_1701497
C##LABCLASSB
NO NO NO NO NO

C##LABCLASSB
C##LABROLENESTED

GRANTEE
-----
GRANTED_ROLE
-----
ADM DEL DEF COM INH
```

```
SQL> DESCRIBE DBA_ROLE_PRIVS
Name                                     Null?   Type
-----
GRANTEE                                VCHAR2(128)
GRANTED_ROLE                           VCHAR2(128)
ADMIN_OPTION                           VCHAR2(3)
DELEGATE_OPTION                         VCHAR2(3)
DEFAULT_ROLE                           VCHAR2(3)
COMMON                                VCHAR2(3)
INHERITED                              VCHAR2(3)

SQL> SELECT *
  2   FROM DBA_ROLE_PRIVS
  3   WHERE GRANTEE LIKE '%STUDENTA%';

GRANTEE
-----
GRANTED_ROLE
-----
ADM DEL DEF COM INH
-----
C##STUDENTA_1701497
C##LABCLASSA
NO NO YES NO NO

SQL>
```

```
SQL> DROP ROLE C##LABCLASSA;

Role dropped.

SQL> SELECT *
  2   FROM DBA_ROLE_PRIVS
  3   WHERE GRANTEE LIKE '%STUDENTA%';

no rows selected

SQL>
```